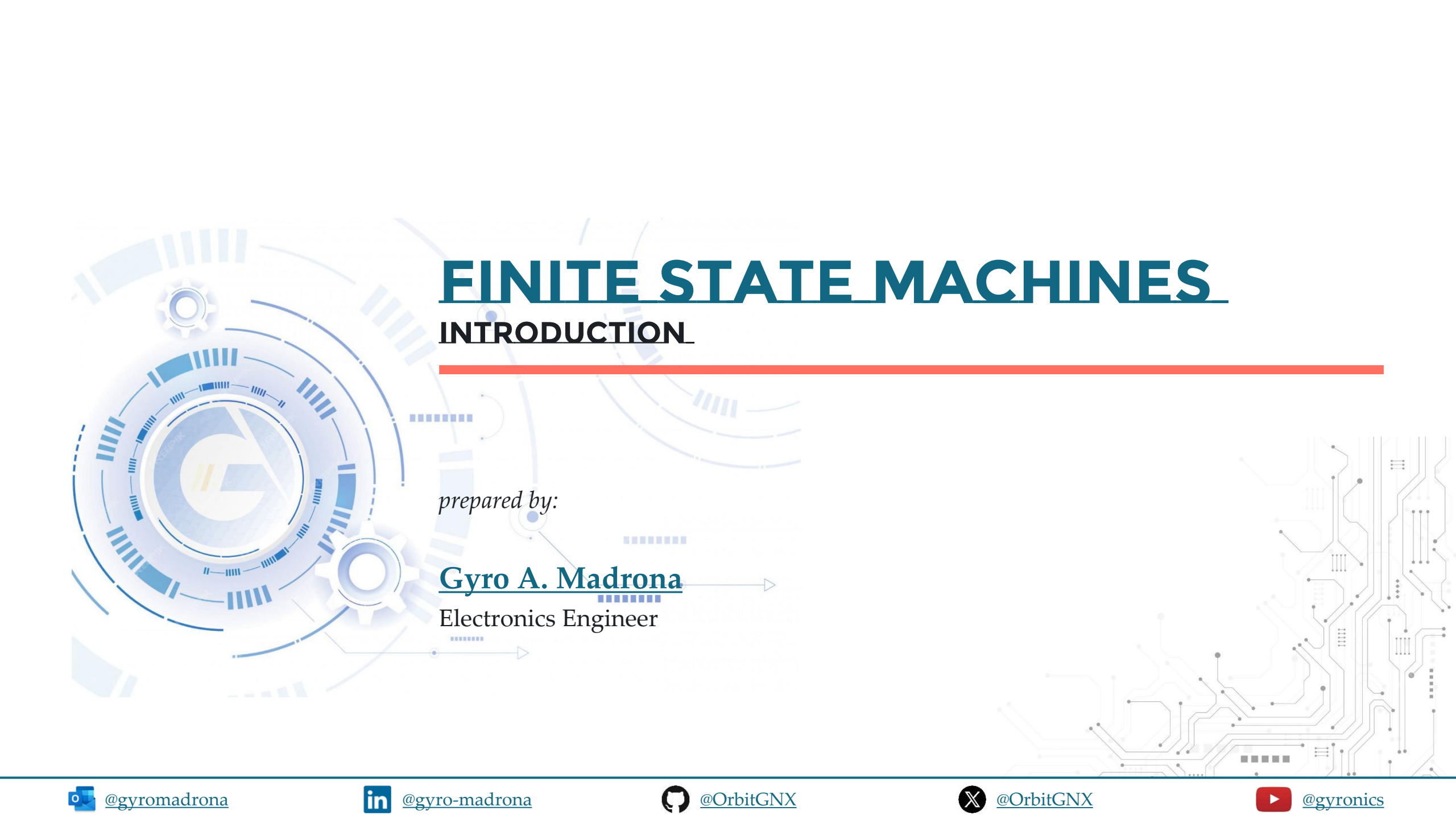




FINITE STATE MACHINES

INTRODUCTION

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Finite State Machines

- Moore Machine
- Mealy Machine



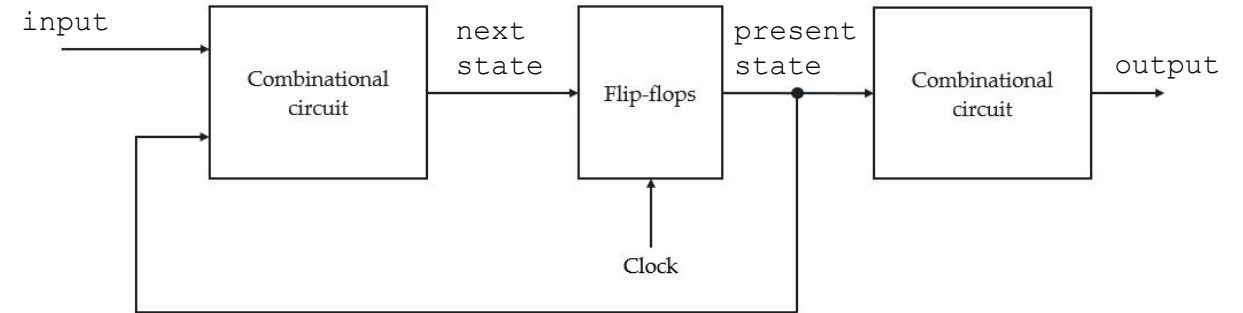
FINITE STATE MACHINES



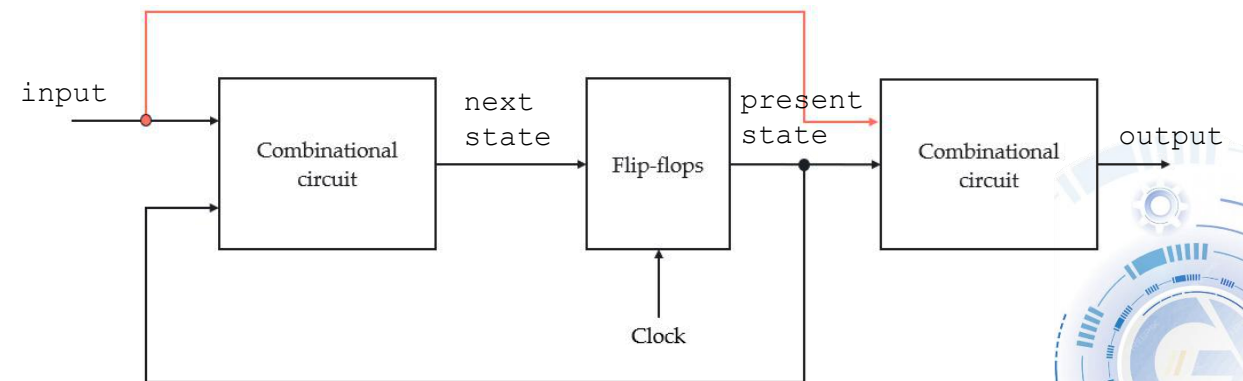
FINITE STATE MACHINES

The name “Finite State Machines” (FSMs) is derived from the fact that the functional behavior of these circuits can be represented using a **finite number of states**.

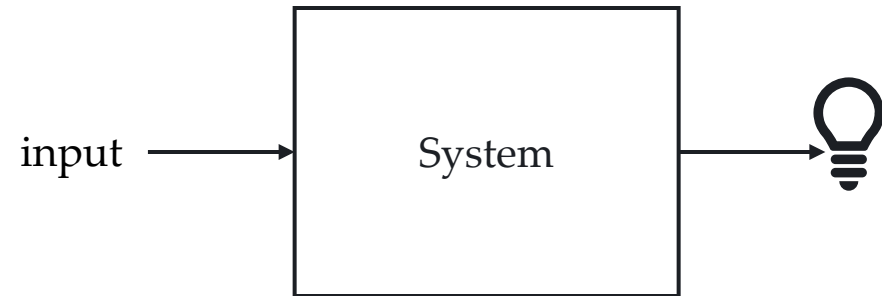
Moore Machine



Mealy Machine



SYSTEM LEVEL DESIGN

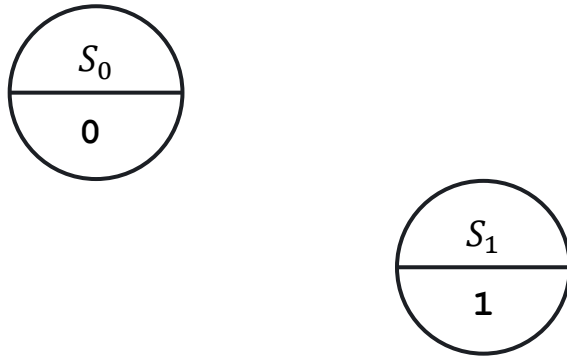


When the input = HIGH

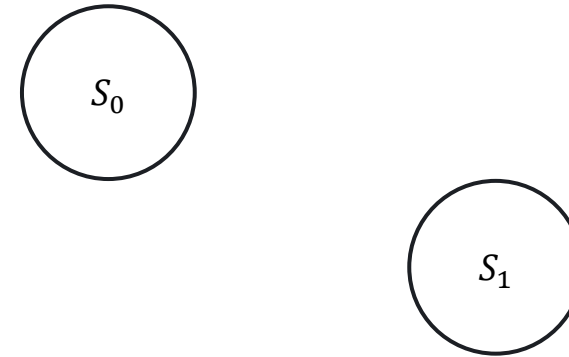
- if the light is OFF, it turns ON
- if the light is ON, it turns OFF

STATE DIAGRAM

Moore Machine

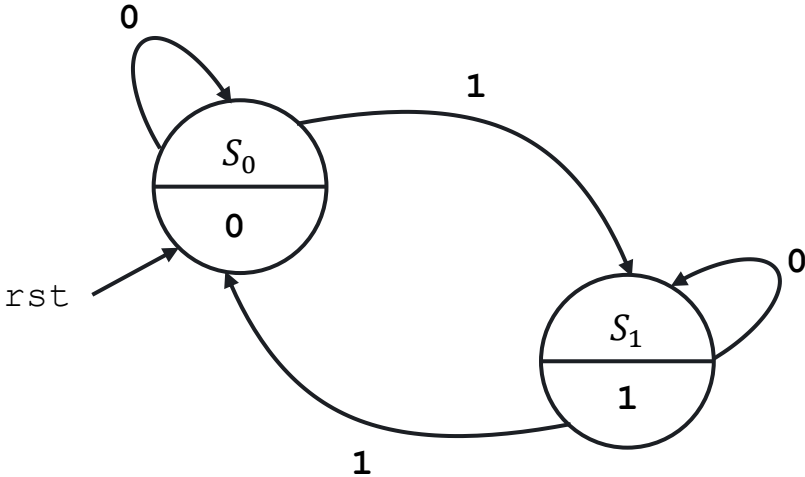


Mealy Machine



STATE TABLE

Moore Machine

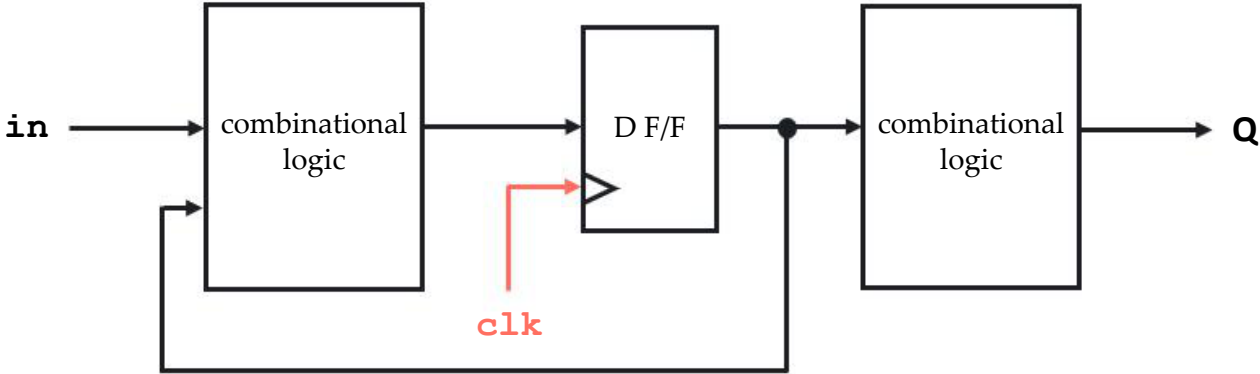


Present State	Next State	Output



STATE ASSIGNMENT

Moore Machine

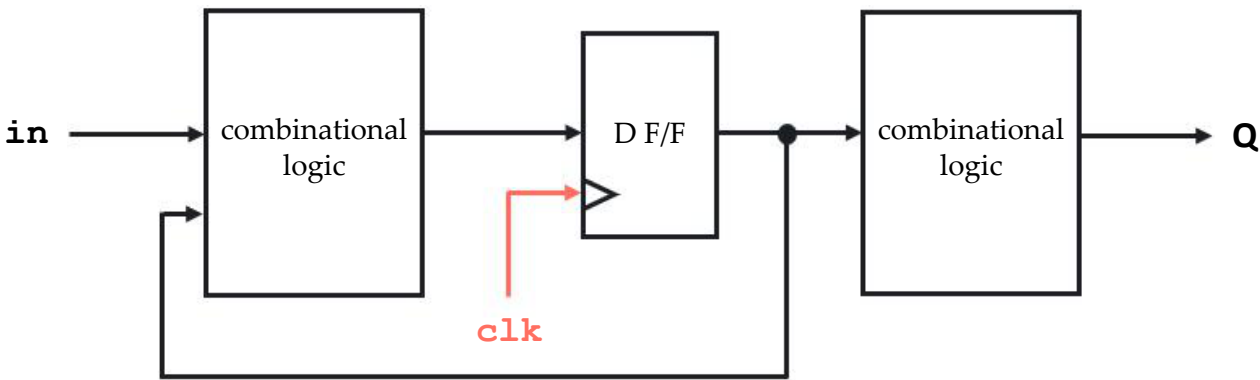


Present State	Next State	Output



LOGIC EXPRESSION

Moore Machine

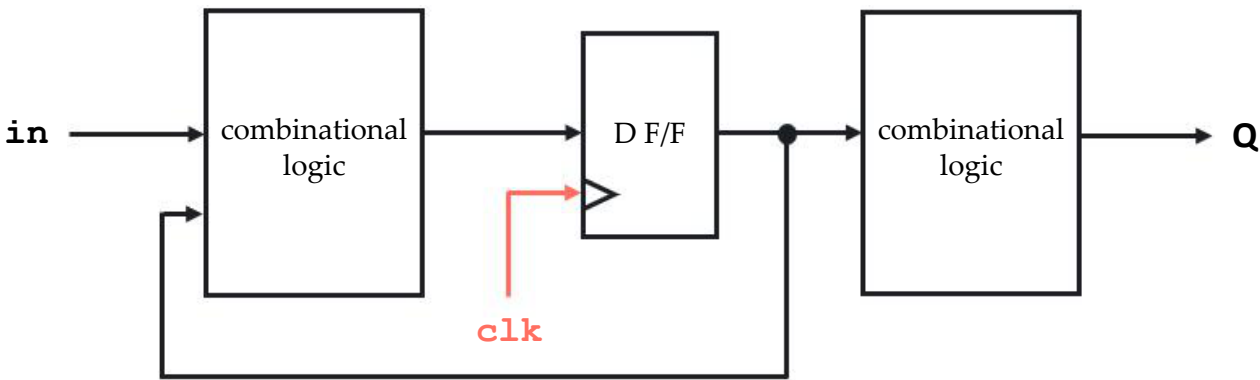


Present State	Next State	Output



LOGIC EXPRESSION

Moore Machine

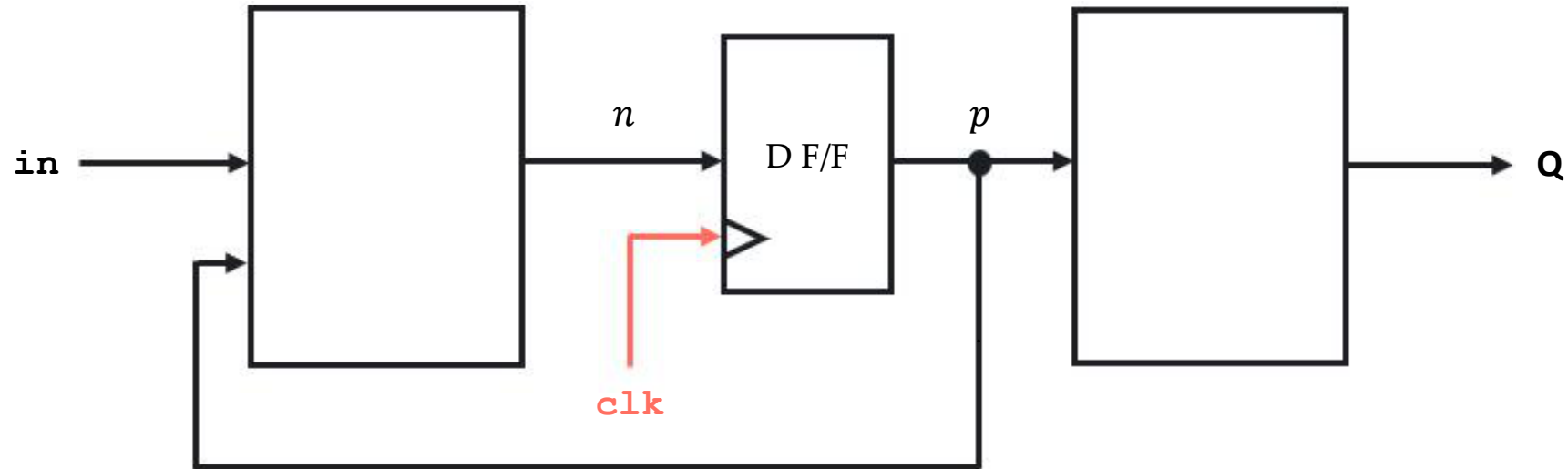


Present State	Next State	Output



IMPLEMENTATION

Moore Machine



LABORATORY

